

DISEASE MODELING & SIMULATION

COPunD : Cellular Automata for COPD Dynamics.

Authors:

Georgy ZAOUK

Bruno YOUNG DE CASTRO

Akshay KHOKAN

Supervised by :

Franck DELAPLACE

Guillaume HUTZLER

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COPunD: Chronic Obstructive Pulmonary underlying Dynamics.

Akshay KHOKAN ¹, Georgy ZAOUK ¹ & Bruno YOUNG DE CASTRO ¹

¹ Master's Student (Year 1) in GENIOMHE-AI, Paris-Saclay University

INTRODUCTION

Chronic Obstructive Pulmonary Disease (COPD) is a chronic inflammatory lung disease mainly caused by long-term exposure to airborne pollutants, particularly cigarette smoke. It is characterized by persistent airflow obstruction, chronic inflammation, and structural remodeling of the small bronchi. Patients may also experience exacerbations, which are acute episodes of worsening respiratory symptoms that contribute to disease progression ([Dupin et al., 2023](#)).

At the cellular level, COPD involves complex interactions between fibrocytes and CD8+ T lymphocytes within lung tissue. Fibrocytes are circulating mesenchymal progenitor cells that migrate into damaged areas, where they contribute to fibrosis and tissue remodeling. In COPD, their infiltration into the lungs is significantly increased compared to healthy conditions. CD8+ T cells are cytotoxic immune cells involved in inflammatory responses and are known to accumulate abnormally in diseased lungs, where they may contribute to chronic inflammation and tissue damage. Their recruitment is thought to be partly influenced by fibrocyte activity ([Dupin et al., 2023](#)).

Because direct experimental observation of long-term cellular interactions in the bronchial environment is difficult due to its structural and cellular complexity, [Dupin et al. \(2023\)](#) proposed a probabilistic cellular automaton model to study these dynamics. Cellular automata are particularly suitable for this purpose, as they model individual cells on a spatial grid and allow complex global spatial patterns to emerge from simple local interaction rules.

In this work, we implement a simplified version of this model in Python, called *COPunD* (*Chronic Obstructive Pulmonary underlying Dynamics*), inspired by [Dupin et al. \(2023\)](#). We compare healthy and COPD-like conditions across different time scales to analyze differences in cell population dynamics and spatial organization. The objective is to investigate whether simple local interaction rules are sufficient to reproduce the distinct spatial patterns observed in healthy versus diseased tissue.

Under healthy conditions, cell populations remain sparse, stable, and spatially homogeneous. In contrast, COPD conditions lead to progressive inflammatory infiltration, increased cell density, and the emergence of spatial clustering driven by local interactions between fibrocytes and CD8+ T cells.

MODELING

Our model is based on the probabilistic cellular automaton proposed by [Dupin et al. \(2023\)](#), but it was simplified to fit the scope of this project and to be implemented using the cellular automata framework provided during the course. Since the introduction already presented the biological background and the general principles of the automaton, this section focuses on the computational implementation of the model. The implementation was built using the cellular automata framework, *cellularautomata.py*, which provides the basic simulation and visualization tools used in this project ([Delaplace, 2024](#)).

1. Cellular Representation and Initialisation

In our simplified version, each site of the lattice can be in one of three states: *Empty*, *Fibrocyte*, or *CD8*. The empty state represents unoccupied tissue space or biological components not explicitly modeled. Fibrocytes represent structural cells involved in tissue remodeling, while CD8 cells represent inflammatory immune cells. The initial cell densities were chosen to approximate the proportions described in the original paper, with a majority of empty sites and smaller populations of fibrocytes and CD8 cells ([Dupin et al., 2023](#); [Khokan et al., 2026](#)).

2. Time-scaling

A time-scaling parameter called $K_TIME \in \{1, 10, 20, 50\}$ was introduced to accelerate biological processes such as infiltration, proliferation, and cell death. This was necessary because COPD is a chronic disease whose effects emerge over long periods of time. By increasing K_TIME , the model allows long-term disease dynamics to be observed within a shorter computational simulation. Event probabilities are rescaled using

$$p_{scaled} = 1 - (1 - p_{base})^k$$

3. Update Rules

The automaton evolves according to local probabilistic rules implemented in the function *CopdPCA()* in *copd_Bruno.py*. It defines the behaviour of each cell based on its type and its Moore neighbourhood (radius 1, eight neighbours). The simulation uses synchronous updates, meaning all cells are updated simultaneously at each step from the previous grid state.

3.1 Empty cells

An empty cell can become a fibrocyte through tissue infiltration (COPD only), or a CD8 cell if at least one fibrocyte is present in its neighbourhood.

The fibrocyte requirement for CD8 birth enforces the biological constraint that CD8 recruitment is driven by fibrocyte-mediated inflammatory signalling. Empty cells can also be filled by a migrating neighbour.

3.2 Fibrocyte cells

A fibrocyte can die with a time-scaled probability, or attempt to migrate. Migration probability is modulated by two functions: *cohesion()*, which reduces mobility in dense same-type clusters, and *cluster_attraction()*, which increases the tendency to aggregate. When migration is attempted, the fibrocyte signals its intent by returning a 'migrating' status.

3.3 CD8 cells

CD8 cells follow the same death and migration logic as fibrocytes, with different base probabilities. CD8 cells are less mobile than fibrocytes.

4. Migration

In the CA library, each cell is updated independently, which makes direct movement between positions impossible. A cell can only update its own state, and there is no built-in mechanism to coordinate swaps with neighbouring cells. To overcome this limitation, we make use of the synchronous update rule, where all cells read from the same previous grid state at the same time. As a result, even if a cell decides to move during an update step, its neighbours still see it in its original position when they compute their own updates. We exploit this feature to simulate movement. A cell that intends to move marks itself as “migrating”. Then, if a neighbouring empty cell detects this marker, it adopts the type of the migrating cell. This creates the effect of a cell shifting to a new position without modifying the library.

The main drawback of this method is that the original cell is not removed immediately. It only disappears in the following step, which can briefly make it look like there are two cells instead of one. This is acknowledged as a simplification of the mobility relative to the original Dupin et al. model., but it remains consistent with the constraints of the system.

5. Condition differences

Two conditions are simulated. In the control condition, fibrocyte infiltration is disabled by setting its probability to zero rather than assigning it a low value. and CD8 birth probability is low. In the COPD condition, fibrocyte infiltration is active and CD8 birth probability is an order of magnitude higher, reflecting the elevated inflammatory signalling characteristic of COPD.

SIMULATION

The following section presents the experimental setup used for the simulations, followed by the results obtained across different time scales and the resulting spatial dynamics observed in each condition.

1. Short-Term Simulation (12 Days)

To compare healthy and COPD-like conditions, we ran the model with the same initial cell densities and tested several biological time scales: 12, 125, 250, and 625 days. The `demo.py` script automatically runs both conditions, saves population curves, and produces initial/final snapshots for each case. The grid size was fixed at 50×50 cells, with 300 simulation steps and initial proportions of approximately 96.5% empty sites, 0.5% fibrocytes, and 3% CD8 cells.

The short simulation of 12 days already shows a difference between the two conditions, but the effect remains moderate. In the control condition, fibrocytes and CD8 cells remain relatively stable or decrease slightly. In the COPD condition, both populations increase, especially CD8 cells. This agrees with the idea that COPD is not an immediate process but a progressive chronic disease. The evolution of the cell populations and the spatial states for the 12-day simulation are shown in **Figures 1 and 5 (Appendix)**.

2. Long-Term Simulations (125–625 Days)

The longer simulations are more informative. For 125, 250, and 625 days, the COPD condition shows a strong increase in fibrocytes and CD8 cells, while the control condition progressively loses cells. The population curves show that the COPD system reaches a much denser inflammatory state, whereas the control system remains sparse. These behaviors can be observed in **Figures 2, 3, and 4 (Appendix)**.

3. Spatial Organization of Cells

The snapshots confirm this difference visually: the control tissue stays almost empty after long simulation times, while the COPD tissue becomes filled with fibrocytes and CD8 cells forming dense spatial patterns. The spatial organization of the tissue for the different simulation durations is presented in **Figures 5 to 8 (Appendix)**.

These results reproduce the main qualitative behavior described by [Dupin et al. \(2023\)](#): local interactions between fibrocytes and CD8+ T cells can generate different tissue organizations in healthy and diseased conditions. In our simplified model, the most important result is that COPD-like rules lead to progressive accumulation of inflammatory cells over time.

This makes the longer simulations more biologically relevant, since COPD develops slowly and its effects are expected to appear more clearly over extended periods rather than after only a few days.

CONCLUSION & LIMITATIONS

This project implemented a simplified probabilistic cellular automaton inspired by [Dupin et al. \(2023\)](#) to model interactions between fibrocytes and CD8+ T lymphocytes in COPD tissues. Despite its simplifications, the model reproduced the progressive accumulation of inflammatory cells under COPD-like conditions. These results show how simple local interaction rules can generate complex spatial behavior over long periods of time.

However, the model remains a simplified version of the original work. Compared with [Dupin et al. \(2023\)](#), it uses fewer biological states, a reduced number of probabilities, a square grid instead of a bronchial geometry, and a less detailed migration mechanism (**Table 1**). In particular, other lung tissue components are grouped into the Empty state, stable and exacerbation-related infiltration are not separated, and migration is represented mainly through mobility and clustering rules rather than detailed chemotactic transition probabilities. Therefore, this model should be interpreted as a qualitative and educational approximation of COPD-related inflammation rather than as a fully calibrated biomedical model.

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APPENDIX

High-resolution versions of all figures referenced in the report are available in the [COPunD repository](#) to ensure full quality and reproducibility of the results.